Year 12 Maths - Probability

Similar questions - Venn diagrams, conditional probability and games

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 60 students were asked whether they study Art (A) and Business (B). The numbers are shown in the Venn diagram: 14 study only Art, 18 study only Business, 9 study both, and 19 study neither.

(a) Draw the Venn diagram, and find $\mathrm{P}(A)$. **[2]**

(b) Find $\mathrm{P}(A \mid B)$. **[2]**

(c) Determine whether A and B are independent events, showing your working. **[3]**

(d) Two different students are chosen at random. Find the probability that both study exactly one of the two subjects. **[3]**

2 Asha and Ben play a game. On each point, Asha wins with probability 0.6, independently of other points. The first player to win 2 points wins the game.

(a) Show that the probability that Asha wins the game is 0.648. **[4]**

(b) Find the probability that the game lasts exactly 3 points. **[3]**

(c) Given that the game lasts exactly 3 points, find the probability that Asha wins. Comment on your answer. **[3]**

Question 1

(a) Venn diagram with 14 in A only, 9 in the overlap, 18 in B only, 19 outside. $\mathrm{P}(A) = \dfrac{14+9}{60} = \dfrac{23}{60}$

(b) $\mathrm{P}(A \mid B) = \dfrac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)} = \dfrac{9/60}{27/60} = \dfrac{1}{3}$

(c) $\mathrm{P}(A)\mathrm{P}(B) = \dfrac{23}{60} \times \dfrac{27}{60} = 0.1725$ but $\mathrm{P}(A \cap B) = \dfrac{9}{60} = 0.15$. These are not equal, so A and B are not independent

(d) 32 students study exactly one subject: $\dfrac{32}{60} \times \dfrac{31}{59} = \dfrac{248}{885} = 0.280$

Question 2

(a) Asha wins in 2 points: $0.6^2 = 0.36$. Asha wins in 3 points (Ben wins one of the first two): $2 \times 0.6 \times 0.4 \times 0.6 = 0.288$. Total $0.36 + 0.288 = 0.648$

(b) Three points are needed if the first two are split: $2 \times 0.6 \times 0.4 = 0.48$

(c) Asha wins in exactly 3: 0.288. $\dfrac{0.288}{0.48} = 0.6$. This equals the probability of Asha winning any single point, as the winner of the third point wins; given a split start, only the last point matters